Vertex Form of a Quadratic Function

Name _____

Algebra 2 Lesson 2-1 (fast track) Date _____ Period _____

Learning goal: I can write the equation of a parabola in vertex form from its vertex and one other point, or from a translation or reflection of $y = x^2$, and rewrite it in the form $y = ax^2 + bx + c$.

Language goal: I can explain how I wrote an equation using the words *vertex*, *translation*, *reflection*, *opens upward / downward* and *maximum / minimum*.

On the quiz: You will (1) tell how $g(x) = (x + 2)^2 + 4$ moves $y = x^2$ (full Savvas quiz only), (2) find a bird's greatest height from vertex form (full Savvas quiz only), (3) write vertex form from the vertex and a point (Example 4), (4) read the vertex and y-intercept off a graph, find a, write $y = ax^2 + bx + c$ (Example 4), (5) move $y = x^2$ right 3 and down 5 and write it as $y = ax^2 + bx + c$ (Example 5).

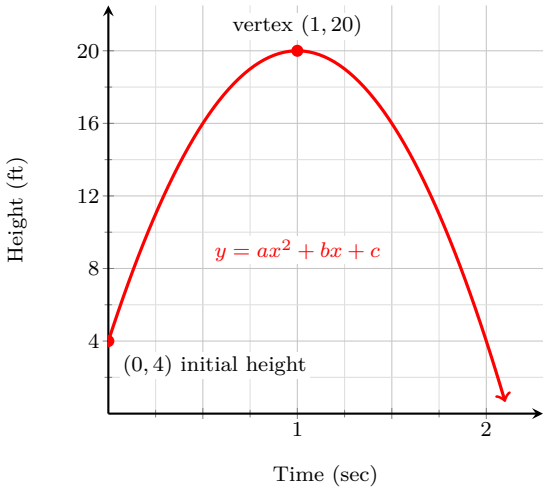
On the Topic 2 test: Question 1 asks you to write the golf ball's distance from the hole over time in vertex form from its vertex and one point (Example 4). Question 2 asks you to reflect $y = x^2$ in the x-axis, move it left and down, and write it as $y = ax^2 + bx + c$ (Example 5).

Example 4: Write an Equation of a Parabola Given the Graph

The height of a hit ball is a quadratic function of the time it has been in the air. The ball is hit with an initial height of 4 feet and after 1 second reaches its vertex at 20 feet. What is an equation that defines this function? Write the quadratic equation in vertex form and in the form $y = ax^2 + bx + c$.

1. The **vertex** is $(h, k) = (1, 20)$, so $y = a(x - 1)^2 + 20$.
2. The initial height is the y-intercept, so $(x, y) = (0, 4)$ is on the graph. Substitute it and solve for a: $4 = a(0 - 1)^2 + 20$, so $4 = a + 20$, $a = -16$
3. Vertex form: $\boxed{y = -16(x - 1)^2 + 20}$
4. Expand to get $y = ax^2 + bx + c$: $y = -16(x^2 - 2x + 1) + 20 = -16x^2 + 32x - 16 + 20$ $\boxed{y = -16x^2 + 32x + 4}$

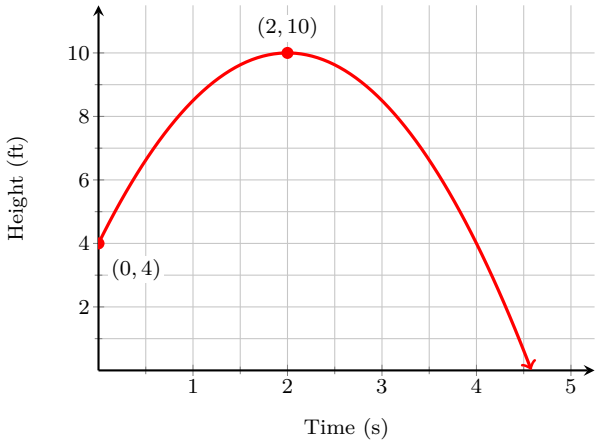
Check: $c = 4$ is the initial height, and $a = -16 < 0$, so the parabola opens downward: the vertex is a maximum.



Say it: The vertex is _____, so $h =$ ____ and $k =$ ____. I substituted the point _____ for x and y and got $a =$ _____. The parabola opens _____ because a is _____, so the vertex is a _____.

Try It 4: Write the equation from the graph

The graph shows the height of a ball with respect to time. What is the equation of the function? Write the equation in vertex form. Then write the equation in the form $y = ax^2 + bx + c$.



Vertex $(h, k) =$ _____
Point $(x, y) =$ _____
$y = a(x - ____)^2 + ____$
Substitute the point and solve for a:

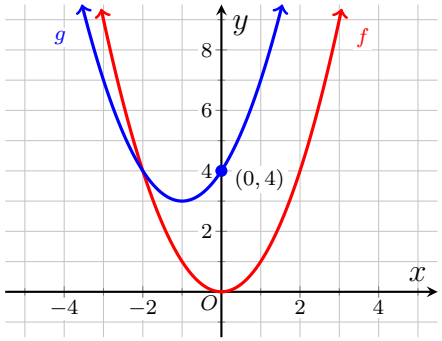
Vertex form: $y =$ _____

Expand to get the form $y = ax^2 + bx + c$:

$y =$ _____ Which number in this form is the initial height? _____

Example 5: Write an Equation of a Transformed Function

The function g is a translation of the parent function f 1 unit left and 3 units up. What is the equation of g? Write the quadratic equation in vertex form and in the form $y = ax^2 + bx + c$.



1. Translate the graph of $f(x)$ left 1 unit to locate the graph of $f(x+1)$, then translate the graph of $f(x+1)$ up 3 units to locate the graph of $f(x+1)+3$. The **vertex** moves from $(0, 0)$ to $(-1, 3)$.
2. $f(x) = x^2$ and $g(x) = f(x+1)+3$, so $g(x) = a(x+1)^2+3$; a is the vertical stretch.
3. From the graph, the point $(0, 4)$ is on g. Use it to find a: $4 = a(0+1)^2+3$, so $4 = a+3$, $a = 1$
4. Vertex form: $\boxed{g(x) = (x+1)^2+3}$
5. Expand to get $y = ax^2 + bx + c$:
   $g(x) = x^2 + 2x + 1 + 3$ $\boxed{g(x) = x^2 + 2x + 4}$

Check: pick a point on the graph and make sure it satisfies your equation.

Say it: The _____ left 1 and up 3 moves the vertex to _____, so $h =$ _____ and $k =$ _____. The point _____ gives $a =$ _____, so g opens _____ and k is the _____ value.

Try It 5: Write the equation of a transformed function

What is the equation of j? Write the equation in vertex form and in the form $y = ax^2 + bx + c$.

a. Let j be a quadratic function whose graph is a translation 2 units right and 5 units down of the graph of the parent function f.

Vertex $(h, k) =$ _____ $a =$ _____
Vertex form: $j(x) =$ _____
Expand:

_____

_____

_____

$j(x) =$ _____

b. Let j be a quadratic function whose graph is a reflection of the graph of the parent function f in the x-axis followed by a translation 1 unit down.
Hint: a reflection in the x-axis makes $a = -1$.

Vertex $(h, k) =$ _____ $a =$ _____
Vertex form: $j(x) =$ _____
Expand:

_____

_____

_____

$j(x) =$ _____

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
URV ★	Write a quadratic function in vertex form from its vertex and another point	Quiz 3 and 4: vertex + one point gives a (Example 4)
KQL ★	Transformations of quadratic functions	Quiz 5: the moves of $y = x^2$ give the vertex and a (Example 5)
WMS	Characteristics of quadratic functions: graphs	Optional warm-up: read the vertex off a drawn parabola (Quiz 4)
L8C	Characteristics of quadratic functions: equations	Optional: Quiz 1 and 2 (full Savvas quiz only)

Extension: Practice 15 (DOK 3), when you finish

Amaya is standing 30 ft from a volleyball net. The net is 8 ft high. Amaya serves the ball. The path of the ball is modeled by the equation $y = -0.02(x-18)^2+12$, where x is the ball's horizontal distance in feet from Amaya's position and y is the distance in feet from the ground to the ball.

a. How far away horizontally is the ball from Amaya when it is at its maximum height? Explain.

The vertex is _____. The parabola opens _____ because $a =$ _____, so the vertex is the _____. The ball is _____ ft from Amaya because

b. Describe how you would find the ball's height when it crosses the net at $x = 30$.

Before the quiz, finish these sentences.

In $y = a(x-h)^2+k$, the vertex is _____. A translation left or right changes _____; a translation up or down changes _____.

A reflection in the x-axis makes a _____, so the parabola opens _____ and k is the _____ value.

To find a, I substitute the _____ for h and k and one other _____ for x and y.